VICTOR MBUGUA

DevOps & Cloud Engineer | Full-Stack Software Developer | ICT Operations

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PROFESSIONAL SUMMARY

Results-driven Software Engineer and certified DevOps practitioner with a BSc in Information Technology and end-to-end experience across the software delivery lifecycle — designing REST APIs, shipping full-stack web and mobile applications, and automating build, test, and deployment pipelines. Recently completed a **DevOps Engineering certification (Moringa School)**, with hands-on command of CI/CD (Jenkins, GitHub Actions), containerization and orchestration (Docker, Kubernetes), and Infrastructure-as-Code (Terraform, Ansible) on cloud platforms. Proven track record running ICT operations for the national Digital Literacy Programme, resolving complex system and network faults, and translating business needs into reliable, secure, and scalable technical solutions.

CORE TECHNICAL SKILLS

DevOps & CI/CD Jenkins, GitHub Actions, GitLab CI, CircleCI, ArgoCD, Maven/Gradle build pipelines

Containers & Orchestration Docker, Docker Compose, Kubernetes, Helm

Infrastructure as Code & Config Terraform, Ansible, AWS CloudFormation, Nginx, Linux administration

Cloud & Monitoring AWS (EC2, S3, IAM, EKS, Lambda), Prometheus, Grafana, ELK Stack

Languages Java, Python, JavaScript / TypeScript, Dart, SQL, Bash, Solidity

Frameworks & Frontend Spring Boot, Django REST Framework, Next.js, React, Angular, Flutter, Node.js

Databases & APIs PostgreSQL, MySQL, SQLite, REST API design, WebSockets, M-Pesa / payment integrations

Practices & Operations Git & version control, Agile SDLC, system analysis & design, testing & debugging, web hosting, LAN/WAN & wireless networking, IT helpdesk support, data security

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

DevOps Engineering Certification — Moringa School

Comprehensive cloud-native delivery toolchain: Linux • Git • Jenkins • GitHub Actions • GitLab CI • Docker • Kubernetes • Helm • Terraform • Ansible • AWS • Prometheus • Grafana • Bash & Python scripting • Nginx

Hackathons & Workshops

KAPs Hackathon and Eclectics Hackathon — Jomo Kenyatta University of Agriculture and Technology (JKUAT).

PROFESSIONAL EXPERIENCE

ICT Officer — Digital Literacy Programme (DLP) — 2024 – 2025

Ministry of Education · Kikuyu Constituency, Kiambu County

- Operated the Constituency ICT Office for 12 months under the national Digital Literacy Programme, serving as the single point of contact for school heads, teachers, and DLP field officers across primary schools in Kikuyu Constituency.
- Designed and launched a teacher-facing request workflow (previously non-existent) for allocating and tracking classroom learning devices — standardizing intake forms, approval routing, and dispatch records to deliver predictable, repeatable turnaround times.
- Provided first-line technical support on DLP devices and supporting infrastructure, diagnosing faults and escalating hardware for repair or replacement to minimize classroom downtime.
- Trained non-technical teaching staff on device care, classroom integration of digital learning content, and the new request workflow, accelerating programme adoption.

Industrial Attaché — Software Development — Feb 2022 – Apr 2022

Tinka Traders Digital Solutions Limited

- Contributed across the full software development life cycle — system analysis, design, development, testing, and debugging — for client web systems.
- Built REST API endpoints for a timetabling application, improving functionality and user interactivity.

- Maintained and updated production websites and managed web hosting, optimizing server performance and uptime.
- Administered databases with best practices for organization, retrieval, and security; developed responsive Angular front-ends aligned with modern UI/UX standards.

IT Officer Nov 2021 – Jan 2022

Stantechmotors International Limited

- Maintained the accuracy and confidentiality of real-time GPS tracking data for the full customer base, safeguarding sensitive information.
- Produced monthly tracking reports that gave a comprehensive view of client data and informed strategic decision-making.
- Diagnosed and resolved complex hardware, software, and network faults, improving overall system and network stability.
- Monitored systems and networks and implemented preventive measures to proactively reduce incidents and downtime.

KEY PROJECTS

- **learnium — Multi-Tenant School Management SaaS —** Kenya-first, multi-tenant platform that unifies the academic record (CBC and 8-4-4), M-Pesa fee collection, and statutory payroll under one login. Architected as six independent Django services (identity and tenancy, academics, finance, payroll, communications, reports), each owning its own PostgreSQL database, behind a Kong API gateway that verifies JWT authentication at the edge and enforces per-tenant role-based access. Services integrate over REST and a Kafka domain-event bus, while Celery and Redis handle M-Pesa payment callbacks, bulk SMS and email, PDF report cards, and scheduled data exports.
  Stack: *Django 6 / DRF microservices, Kong gateway, PostgreSQL (per service), Kafka event bus, Celery + Redis, Next.js 16 / React 19, Flutter, Docker, Kubernetes, GitHub Actions CI*
- **PurePocket (tidyAI) — Offline-First Finance App with On-Device AI —** Local-first Flutter application that reads M-Pesa and bank transaction messages straight from a phone's SMS and notification streams, parses and de-duplicates them into a single encrypted ledger, and runs categorization and spending insights entirely on the device with no server or account. Data is held with AES-256 encryption at rest, a background worker keeps the ledger current, and an on-device language model phrases insights with no data leaving the phone.
  Stack: *Flutter / Dart, Riverpod, Drift + SQLCipher (AES-256), TensorFlow Lite classifier, on-device LLM, biometric app lock, background workers*
- **MyPA — AI Personal Assistant —** Full-stack AI assistant developed for MyPA that unifies personal productivity (task management, scheduling, and reminders), natural-language conversational Q&A, and customer-support and business workflow automation — accessible through chat, voice, and WhatsApp. Delivered end-to-end across web, mobile, and LLM-powered backend services, including prompt design, retrieval-augmented generation, and third-party integrations.
  Stack: *Python (FastAPI / Django) + LLM APIs with RAG & vector search, Next.js / React web, Flutter mobile, Node.js / TypeScript services, WebSockets*
- **Xuperb —** End-to-end car-rental and garage management platform with multi-role portals for admins, agents, drivers, and clients — featuring digital contracts with e-signatures, fleet tracking, invoicing, and a loyalty programme.
  Stack: *Django REST Framework, Next.js 16 + React 19 dashboard, Flutter mobile app*
- **AssureHub —** Audit and consulting management platform unifying workspace management, compliance tracking, and automated reporting, with dual audit/consulting workflows.
  Stack: *Next.js 15, Django REST Framework, multi-factor authentication, role-based access control*
- **SpinWish —** Real-time DJ-and-listener platform for live music sessions, song requests, and tip-based payments.
  Stack: *Spring Boot 3 / Java 17, WebSocket gateways, PostgreSQL, Flutter, M-Pesa & PayMe integration*
- **Blockchain Anti-Counterfeit System —** Decentralized web application tracking product supply chains and guaranteeing item uniqueness via smart contracts.
  Stack: *Ethereum, Solidity smart contracts, decentralized (dApp) architecture*
- **Tourism Destination App —** Mobile app surfacing Kenyan tourist information to aid trip planning and destination discovery.
  Stack: *Java, Android Studio SDK*

EDUCATION

BSc, Information Technology 2019 – 2022

Jomo Kenyatta University of Agriculture and Technology (JKUAT)

Kenya Certificate of Secondary Education (KCSE) 2011 – 2015

Thika High School

REFEREES

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